Sparsh Goswami

AI & Software Engineer | Full Stack Developer | Software Engineer

+91-8956579969 | 1774.sparsh@gmail.com | Nagpur, India

LinkedIn | GitHub | Portfolio

SUMMARY

AI and software engineering student with hands-on experience building RAG pipelines, agentic AI backends, computer vision systems, and full-stack web applications using Python, FastAPI, PostgreSQL, and React. Co-inventor on two published Indian patents (embedded safety automation and blockchain-based procurement) and Top 30 national finalist at the Union Bank of India IDEA 2.0 Hackathon. Also experienced in ESP32-based embedded and IoT systems. Event Manager at the IEEE Student Branch, SIT Nagpur, seeking an AI/software engineering internship.

EDUCATION

Symbiosis Institute of Technology, Nagpur — Nagpur, Maharashtra

B.Tech in Computer Science and Engineering (Specialization in AI & ML) — 2024 – Present

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Embedded Systems

EXPERIENCE

Artificial Intelligence Intern — May 2026 – Jun 2026

CodeAlpha — Remote

- Built a real-time **retail analytics system** using **YOLOv8 with SORT and DeepSORT** to detect and track people, measuring footfall, dwell time, and boundary transitions
- Developed a Tkinter dashboard with live counts and an automated video simulator mode, enabling evaluation without camera hardware
- Built **Linguify**, a Streamlit translation app supporting **90+ languages** with automatic source detection, text-to-speech, text file ingestion, and session history restore

Web Development Intern — Completed

Elevate Labs — Remote

- Built responsive, mobile-first web interfaces using **HTML5, CSS3, and JavaScript**, refactoring layout components for cross-browser stability and accessibility
- Followed **Git and GitHub** team workflows including feature branches, pull requests, and code reviews

PROJECTS

CustomerPulse | *Python, FastAPI, PostgreSQL, pgvector, Amazon Bedrock, S3, Athena* — GitHub | 2026

- Built the **data ingestion layer** importing complaint data from Amazon S3 via Athena into PostgreSQL, plus email and web intake channels feeding the triage pipeline
- Implemented the **analytics and export/reporting modules**; the platform uses a **pgvector RAG pipeline** with Amazon Bedrock to retrieve similar past complaints and draft resolutions
- Selected as a **Top 30 national finalist** at the Union Bank of India IDEA 2.0 Hackathon

Nivaas | *Python, FastAPI, Groq, PostgreSQL, Tavily, Scikit-learn* — GitHub | 2026

- Built an **agentic property recommendation** backend that converts ambiguous natural-language requirements into structured SQL filters and vector queries using Groq
- Integrated Tavily web search, RapidFuzz locality matching, and NumPy/Pandas/Scikit-learn pipelines for multi-criteria ranking, property comparison, and EMI calculation

TaskFlow | *React, FastAPI, SQLAlchemy, PostgreSQL, Git* — GitHub | 2026

- Built a **work verification platform** that inspects Git commits and diff hunks against issue acceptance criteria to flag unverified progress and no-op commits
- Persisted evaluation state with SQLAlchemy and PostgreSQL; deployed on Vercel

TenderBlock | *React, Node.js, MySQL, Flow Blockchain, Cadence, IPFS* — GitHub | 2026

- Co-developed a **decentralized government tender platform** on Flow blockchain with on-chain reverse auction smart contracts and non-transferable NFT award certificates
- Anchored tender documents on IPFS and built an event-driven relational indexer for fast web queries; **patent published** (App. No. 202621072831 A)

MemoryLens | *React, FastAPI, FAISS, Python* — GitHub | 2026

- Co-developed an **AI-powered local search assistant** for natural-language document retrieval (e.g., “the budget file from last quarter”) using FAISS vector search
- Implemented **face-based photo matching** from a single reference photo, with dedicated indexing and search pipelines behind a React frontend

Smart Gas Detecting Rover | *ESP32, Embedded C/C++, Gas and Flame Sensors* — 2025

- Built an autonomous **mobile robotic rover** with gas and flame sensing, safety automation, alerts, and live monitoring for industrial and household hazard detection; **patent published** (App. No. 202521125538 A)

TECHNICAL SKILLS

Languages : Python, C, JavaScript, TypeScript, SQL, HTML/CSS
AI & ML : RAG, Generative AI, AI Agents, Vector Embeddings (FAISS, pgvector), Computer Vision (OpenCV, YOLOv8), Scikit-learn
Web & Backend : FastAPI, React, Next.js, Node.js, REST APIs, Streamlit
Databases : PostgreSQL, MySQL, MongoDB, SQLAlchemy
Cloud & Web3 : Amazon Bedrock, Amazon S3, Athena, Flow Blockchain, Cadence, Smart Contracts, IPFS
Embedded : ESP32, Embedded C/C++
Tools : Git, GitHub, Linux, VS Code, Vercel, Postman
Core Competencies : Data Structures & Algorithms, Object-Oriented Programming, Embedded Systems

PATENTS (INDIA, CO-INVENTOR)

Blockchain-Based Government Tender Management System Using Smart Contracts and NFT Award Certificates | App. No. 202621072831 A | Patent Office Journal No. 31/2026, 31/07/2026
System and Method for Autonomous Hazard Detection and Safety Automation Using Mobile Robotic Platform | App. No. 202521125538 A | Patent Office Journal No. 1/2026, 02/01/2026

CERTIFICATIONS

Mastering Agentic AI: From Prompt to Protocols to Production – Udemy (Aug 2026)
Data Structures and Algorithms using Python (Part 1) – Infosys Springboard (Oct 2025)
Introduction to Artificial Intelligence – Infosys Springboard (Jun 2025)
Introduction to Python Programming (AD141) – Red Hat (Mar 2025)
Red Hat System Administration I (RH124 - RHA) - Ver. 10 – Red Hat

VOLUNTEERING, LEADERSHIP & HACKATHONS

Event Manager, IEEE Student Branch, SIT Nagpur (Jul 2026 – Present): organizing technical seminars, hackathons, and workshops
Co-Coordinator, SMARTECH (Robotics) Club, SIT Nagpur (Aug 2025 – Aug 2026): ran workshops on embedded systems and IoT
Sponsorship, Enthusia 5.0 (College Cultural & Technical Fest) – Symbiosis Institute of Technology
PR Team, Hack4Brahma & Hack4Maha Hackathons
Cadet, NCC (Naval Unit)
Top 30 Finalist, Union Bank of India IDEA 2.0 National Hackathon
Participated in hackathons: **IDEA 2.0, CodeHunt, HackAShastra**